

ZrC-ARC OVERVIEW

The following document overviews the general objectives of ZrC-ARC (Zirconium Carbide Architected Refractory Coatings).

THE PROBLEM

Hypersonic flight and advanced space propulsion are advancements both held back by the same fundamental materials challenge. For hydrogen-fueled scramjet engines applicable to hypersonic flight, hot reactive hydrogen gas at temperatures between 1500K and 2500K continuously attacks combustion chamber walls and nozzles, leading to surface cracking, delamination, and structural degradation. These failure modes similarly halted NASA's NERVA Nuclear Thermal Propulsion (NTP) program in the 1970s, where flow channels subjected to corrosive hydrogen gas at similar temperatures proved impossible to protect with the coatings of the time. Decades later, the problem remains unsolved at the fundamental level. ZrC-ARC aims to target these core issues directly, with immediate relevance to hypersonic propulsion and broader implications for systems where refractory coatings must survive sustained exposure to reactive hydrogen gas, such as NTP technology.

OUR TAKE

ZrC-ARC investigates an often overlooked aspect of coating structures: granular morphologies. **ZrC-ARC aims to investigate how the size, shape, and alignment of individual grains within coating structures affects the thermomechanical properties of refractory barriers.** We are targeting the failure mechanisms that lead to coating delamination, namely CTE (coefficient of thermal expansion) mismatches and carbon vacancies (missing carbon atoms within the ZrC lattice structure) that lead to hydrogen trapping and further degradation of the coating.

From a glance, our approach involves a simulation and physical implementation stage. Our teams will work under the guidance of UCI's **Prof. Daniel R. Mumm**, a decorated materials science researcher with direct expertise in thermal spray coatings, hot corrosion, and coating microstructure effects on durability. Evaluation is conducted at up to 1900K via thermal cycling, with characterization via SEM/XRD imaging at **UC Irvine's Materials Research Institute (IMRI)**. A machine learning pipeline assists in image characterization and microstructural analysis, providing quantitative insight at scale that manual analysis would miss. Our final objective is to develop working physical prototypes.

- **STAGE I — LAMMPS & COMSOL MULTIPHYSICS SIMULATIONS**
- **STAGE II — PHYSICAL PROTOTYPING, TESTING, & ANALYSIS**

RECOGNITION

ZrC-ARC received recognition at a national-level proposal writing competition, winning a **\$10,000 seed grant from the NASA L'SPACE Program** in Spring 2026. Our proposal was evaluated against other university-level teams around the country, a competitive environment that validates our work on the grounds of scientific merit and practical potential.

OUR TEAM

Our team consists of highly-capable students at the **University of California: Berkeley, Los Angeles, and Irvine** campuses studying various subjects in materials science, engineering, computer science, mathematics, etc.. We are currently leveraging connections within our communities to secure further support from faculty with expert knowledge on the subject matter, evaluation and imaging equipment, and additional monetary contributions through established foundations.

HOW YOU CAN BE A PART OF THIS

The success of our work is dependent on who is involved. We are seeking partners in the following four areas:

- **Lab Equipment and Access** — Access to CVD & Other Deposition Equipment, Thermal Cycling Furnaces, XRD/SEM/EDS (Imaging Characterization Equipment), etc., is our most urgent need.
- **Faculty and SME Collaboration** — We are looking to talk to individuals with experience in refractory materials, ceramic coatings, hypersonic flight, or nuclear thermal propulsion who are open to advisory and co-investigation roles.
- **Additional Funding** — Our seed grant covers initial testing and equipment. However, full execution would require supplemental support.
- **Industry Partnerships** — Organizations that are working on aerospace materials, hypersonic flight, nuclear propulsion, advanced manufacturing, or anything else aerospace related will likely find overlap with our project.

Please reach out to any of the following members through phone or email if you would like to assist us in our research, and we will get back to you promptly. We sincerely thank you for your interest.

CONTACT

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